

PRD Autopilot

AI-Powered Founder Platform

Turn ideas into engineering-ready PRDs, validation reports, and investor pitch decks in minutes.

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Team Members

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Short Description

PRD Autopilot is an AI-powered founder platform that helps users validate ideas, generate Product Requirements Documents (PRDs), analyze product quality, create investor-ready pitch decks, and collaborate with AI throughout the product development process.

Project Highlights

Idea Validation → PRD Generation → Health Scoring → Backlog Creation → Pitch Decks → AI Chat

- Multi-agent AI PRD generation
- AI-powered idea validation reports
- PRD health scoring and gap analysis
- Ticket and backlog generation
- Investor-ready pitch deck generation
- AI product advisor chat
- PDF, Markdown, and Notion export

Inspiration

Many founders, students, product managers, and startup teams have strong ideas but struggle to turn them into structured product plans. Researching competitors, validating markets, writing PRDs, and creating pitch decks can take days or weeks.

We wanted to build a platform that acts like an AI co-founder, helping users move from a rough idea to a validated, investor-ready product strategy in minutes.

What It Does

PRD Autopilot helps users:

- Validate startup and product ideas using AI-powered market research.
- Generate complete Product Requirements Documents through an AI interview process.
- Score PRD quality and identify gaps.
- Create structured backlogs and product tickets.
- Generate investor-ready pitch decks.
- Chat with an AI product advisor for guidance and iteration.

- Export work to PDF, Markdown, and Notion.

The platform transforms a rough concept into a complete product planning package.

How We Built It

PRD Autopilot was built using Next.js, TypeScript, Tailwind CSS, Supabase, Groq, Vercel, Novus Analytics, and the Notion API.

The platform combines multiple AI-powered workflows, including idea validation, PRD generation, PRD health scoring, backlog generation, pitch deck creation, and AI-assisted product guidance.

User authentication, dashboards, generated documents, validation reports, pitch decks, and project data are handled through Supabase. AI-generated content is powered through Groq and specialized prompt pipelines designed for different parts of the founder workflow.

Challenges We Ran Into

One of the biggest challenges was generating structured AI outputs that remained consistent across many different product categories and industries. We had to make sure the AI could produce useful PRDs, validation reports, pitch decks, personas, acceptance criteria, and success metrics without returning broken or overly generic content.

We also ran into challenges with authentication, session persistence, protected routes, AI rate limits, PDF and Notion exports, and making sure generated content stayed connected to the correct user account.

Another challenge was testing the platform across many different ideas. Some outputs were strong, while others revealed issues such as incomplete validation sections, generic edge cases, or titles being pulled from interview answers instead of generated as proper product names.

Accomplishments We Are Proud Of

- Built a complete AI-powered founder workflow from idea validation to pitch deck generation.
- Created a PRD generation system that turns a rough idea and five follow-up questions into a structured product document.
- Added PRD health scoring and gap analysis.
- Built AI-powered idea validation reports with market, competitor, risk, and SWOT analysis.
- Generated investor-ready pitch decks from ideas, PRDs, or validation reports.
- Added export support for PDF, Markdown, and Notion.
- Built authentication, protected routes, user dashboards, and saved project history.
- Deployed the app to Vercel and tested it across desktop and mobile.

What We Learned

We learned how important prompt engineering, structured AI responses, and real user testing are when building AI-powered tools. Even when the app worked technically, testing showed us where AI output could be improved, where UX could be clearer, and where edge cases could break the experience.

We also learned how to combine multiple AI workflows into one platform. Instead of only generating a document, PRD Autopilot supports a larger product development flow: validating an idea, generating a PRD, scoring the PRD, creating tickets, generating a pitch deck, and chatting with AI for iteration.

What's Next

- Team collaboration and shared workspaces.
- More advanced PRD editing workflows.
- Google OAuth support.
- Integrations with tools like Jira, Linear, Trello, and Slack.
- More customizable pitch decks.
- Stronger validation reports with deeper market intelligence.
- Better AI coaching for product strategy and founder decision-making.
- Enterprise sharing, permissions, and stakeholder feedback workflows.

Tech Stack

Frontend		Next.js 14, TypeScript, Tailwind CSS
Backend	and	Supabase
Database		
AI		Groq, Llama 3.3 70B
Analytics		Novus Analytics
Deployment		Vercel
Integrations		Notion API

Links

- **GitHub Repository:** <https://github.com/erfwn81/PRD-Autopilot>
- **Live Demo:** <https://prd-autopilot.vercel.app>
- **Demo Video:** <https://youtu.be/I_RGj – ewkVc>

Team Members

Name	Responsibilities
Erfan Mirzaee	Backend, AI systems, architecture, deployment
Elijah Rodriguez	Frontend, dashboard, UI/UX, mobile polish
Esmeralda Amado	Authentication, QA testing, demo video, Devpost submission